

UNIT 2 · PART 1

Introduction to LLM Strategies & Chain-of-Thought Reasoning

Large Language Models are powerful — but solving complex problems requires more than generating text. This unit introduces **reasoning strategies** that help models think systematically, reduce errors, and tackle multi-step challenges.



Why Do LLMs Need Reasoning?

Simple questions need no reasoning. But multi-step problems demand it.

✗ No Reasoning Needed

"What is the capital of India?"

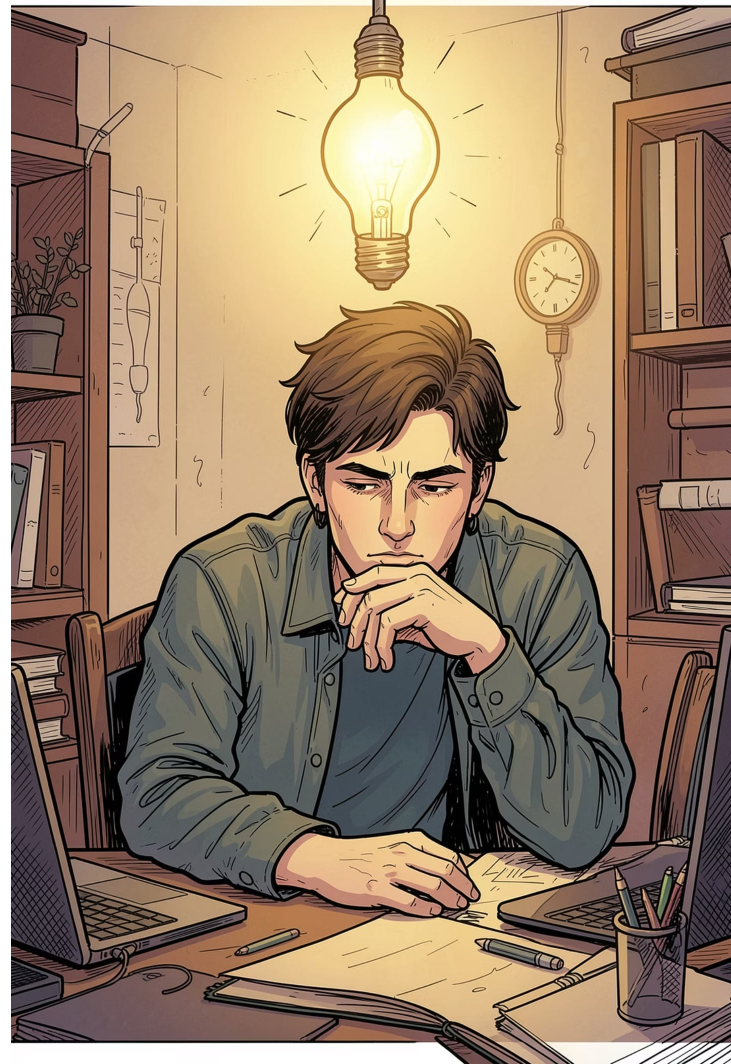
→ **New Delhi**. Direct retrieval.

✓ Reasoning Required

"Rahul is older than Aman. Aman is older than Neha. Who is oldest?"

→ Model must **think through relationships** first.

📌 **Key takeaway:** Reasoning enables LLMs to **solve problems**, not just retrieve facts.



Prompting vs. Reasoning

Prompting

User asks → model responds directly.

"Translate 'Hello' into Hindi."

→ नमस्ते

- Direct answer
- Fast, no intermediate steps
- Suited for simple tasks

Reasoning

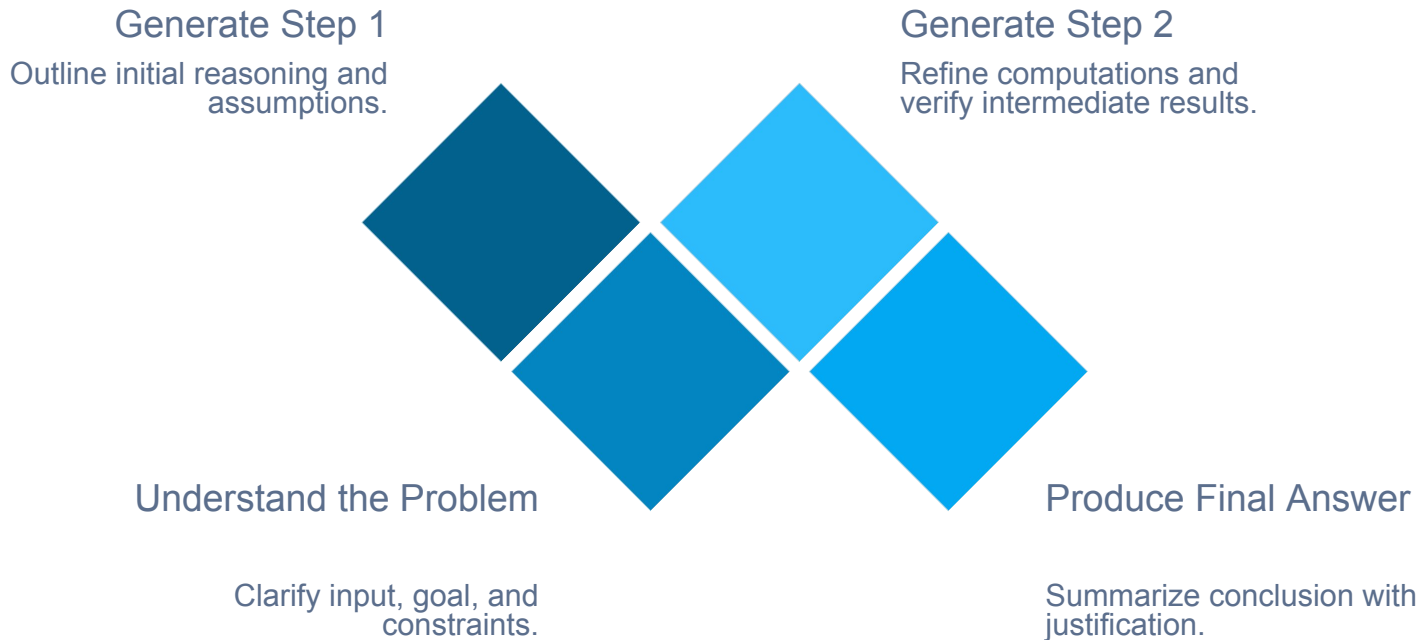
Model breaks the problem into logical steps before answering.

"A train travels 60 km/h. How far in 4 hours?"

- Speed = 60 km/h
- Time = 4 hours
- Distance = Speed × Time
- → **240 km**

What Is Chain-of-Thought (CoT)?

Chain-of-Thought is a reasoning strategy where the LLM generates **intermediate steps** before arriving at a final answer — making its logic transparent and verifiable.



CoT improves mathematical performance, logical reasoning, and decision-making across domains like coding, planning, and research.

CoT in Action: Mathematics



The Problem

A shop sells one notebook for ₹50. A student buys 4 notebooks. How much will the student pay?

Without CoT

Answer: ₹200 — correct, but unverified.

With CoT

01

Cost of one notebook = ₹50

02

Number of notebooks = 4

03

Total = $50 \times 4 = \text{₹}200$

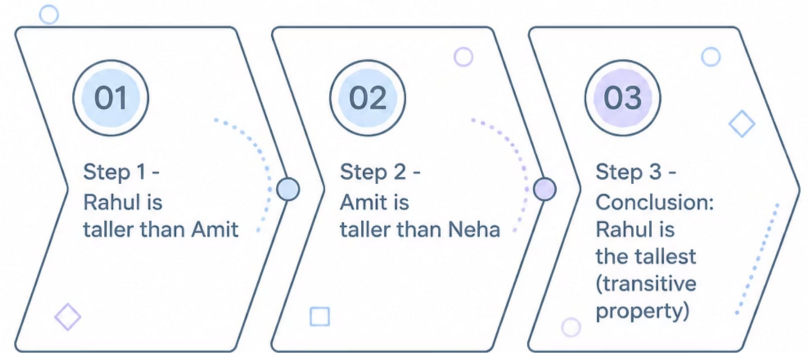
CoT in Action: Logical Reasoning

The Problem

Rahul is taller than Amit. Amit is taller than Neha. Who is the tallest?

CoT Reasoning Steps

Final Answer: Rahul is the tallest.



Rahul > Amit

Amit > Neha

Therefore: **Rahul > Amit > Neha**



Benefits of Chain-of-Thought



Improved Accuracy

Reduces logical errors by forcing structured, step-by-step thinking before committing to an answer.



Better Explainability

Intermediate steps make the model's reasoning transparent and easier to audit or debug.



Handles Complexity

Enables LLMs to tackle multi-step problems in math, coding, planning, and scientific analysis.



Reliable Planning

Supports structured decision-making in medical, financial, and engineering contexts.

Limitations of Chain-of-Thought



Trade-offs to Consider

- **Slower responses** — additional reasoning steps add latency.
- **Higher token usage** — more text generated means higher compute cost.
- **Wrong assumptions** — flawed premises still produce flawed conclusions.
- **Overkill for simple tasks** — "What is 2 + 2?" needs no CoT.



Rule of thumb: Use CoT when the task requires **multiple logical steps**.

When to Use Chain-of-Thought

✓ Best Use Cases

- Mathematics & arithmetic problems
- Programming & code debugging
- Multi-step planning & strategy
- Data analysis & research
- Logic puzzles & decision support

✗ Avoid CoT For

- Simple greetings
- Basic translations
- Single-word factual lookups
- Very simple arithmetic

Classroom Activity

"A farmer has 12 cows. Four are sold, three purchased. How many remain?"

Solve step-by-step, then compare with a direct answer.

Summary & What's Next

Key Concepts

- Reasoning vs. prompting
- CoT workflow & examples
- Benefits & limitations
- When to apply CoT

Core Takeaways

- LLMs need reasoning for complex tasks
- CoT breaks problems into verifiable steps
- Best for math, coding, planning, and analysis

Next: Unit 2 – Part 2

- **Self-Consistency** — multiple reasoning paths
- **Tree-of-Thought** — branching solution exploration
- Foundation for advanced AI agents

❏ These strategies extend CoT and form the backbone of **modern AI agent architectures**.